

Project Performance Testing Phase

Model Performance Testing

Metric	Details
Date	09 July 2026
Team ID	K. Bhargav
Project Name	Predictive Risk Clustering and Visual Informatics for Cardiovascular Disease
Maximum Marks	4 Marks

Performance Testing Log

S.No.	Parameter	Screenshot / Values
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1. Data Rendered

The clinical dataset ([Heart_new2.csv](#)) was successfully uploaded and rendered in Tableau Public. All 18 dataset fields—including target variables like HeartDisease, BMI, Smoking, AlcoholDrinking, Stroke, AgeCategory, and Diabetic status—were loaded correctly and made available for cross-variable analysis without missing row errors.

Tableau Public - Book1

FileDataWindowHelp

Connections

Heart_new2

Files

Use Data Interpreter

Data Interpreter might be able to clean your Text file workbook.

Heart_new2.csv

New Union

New Table Extension

Heart_new2

Heart_new2.csv

Heart_new2.csv

18 fields 4494 rows

100 rows

NameHeart_new2.csv

Description

No description available.

Fields

Type	Field Name	Physical Table	Remote Fi...
Abc	Heart Disease	Heart_new2.csv	HeartDisease
#	BMI	Heart_new2.csv	BMI
Abc	Smoking	Heart_new2.csv	Smoking

AbcHeart_new2.csvHeart Disease

#Heart_new2.csvBMI

AbcHeart_new2.csvSmoking

AbcHeart_new2.csvAlcohol Drinking

AbcHeart_new2.csvStroke

No16.6000YesNoNoNo

No20.3400NoNoNoYes

No26.5800YesNoNoNo

No24.2100NoNoNoNo

No23.7100NoNoNoNo

Yes28.8700YesNoNoNo

(System successfully verified 4,494 complete patient records.)

2.Data Preprocessing

Removed null values, verified strict computational data types (mapping String formats for categorical fields and Integer/Float formats for continuous variables like BMI), created essential calculated fields for patient volume aggregation, grouped Age Categories (18-24 to 80+), and validated boolean condition fields (Yes/No) before visualization.

HeartDisease																		
A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
HeartDisease	BMI	Smoking	AlcoholDrinking	Stroke	PhysicalHealth	MentalHealth	DiffWalking	Sex	AgeCategory	Race	Diabetic	PhysicalActivity	GenHealth	SleepTime	Asthma	KidneyDisease	SkinCancer	
No	16.6	Yes	No	No	3	30	No	Female	55-59	White	Yes	Yes	Very good	5	Yes	No	Yes	
No	20.34	No	No	Yes	0	0	No	Female	80 or older	White	No	Yes	Very good	7	No	No	No	
No	26.58	Yes	No	No	20	30	No	Male	65-69	White	Yes	Yes	Fair	8	Yes	No	No	
No	24.21	No	No	No	0	0	No	Female	75-79	White	No	No	Good	6	No	No	Yes	
No	23.71	No	No	No	28	0	Yes	Female	40-44	White	No	Yes	Very good	8	No	No	No	
Yes	28.87	Yes	No	No	6	0	Yes	Female	75-79	Black	No	No	Fair	12	No	No	No	
No	21.63	No	No	No	15	0	No	Female	70-74	White	No	Yes	Fair	4	Yes	No	Yes	
No	31.64	Yes	No	No	5	0	Yes	Female	80 or older	White	Yes	No	Good	9	Yes	No	No	
No	26.45	No	No	No	0	0	No	Female	80 or older	White	No, border	No	Fair	5	No	Yes	No	
No	40.69	No	No	No	0	0	Yes	Male	65-69	White	No	Yes	Good	10	No	No	No	
Yes	34.3	Yes	No	No	30	0	Yes	Male	60-64	White	Yes	No	Poor	15	Yes	No	No	
No	28.71	Yes	No	No	0	0	No	Female	55-59	White	No	Yes	Very good	5	No	No	No	
No	28.37	Yes	No	No	0	0	Yes	Male	75-79	White	Yes	Yes	Very good	8	No	No	No	
No	28.15	No	No	No	7	0	Yes	Female	80 or older	White	No	No	Good	7	No	No	No	
No	29.29	Yes	No	No	0	30	Yes	Female	60-64	White	No	No	Good	5	No	No	No	
No	29.18	No	No	No	1	0	No	Female	50-54	White	No	Yes	Very good	6	No	No	No	
No	26.26	No	No	No	5	2	No	Female	70-74	White	No	No	Very good	10	No	No	No	
No	22.59	Yes	No	No	0	30	Yes	Male	70-74	White	No, border	Yes	Good	8	No	No	No	
No	29.86	Yes	No	No	0	0	Yes	Female	75-79	Black	Yes	No	Fair	5	No	Yes	No	
No	18.13	No	No	No	0	0	No	Male	80 or older	White	No	Yes	Excellent	8	No	No	Yes	
No	21.16	No	No	No	0	0	No	Female	80 or older	Black	No, border	No	Good	8	No	No	No	
No	28.9	No	No	No	2	5	No	Female	70-74	White	Yes	No	Very good	7	No	No	No	
No	26.17	Yes	No	No	0	15	No	Female	45-49	White	No	Yes	Very good	6	No	No	No	
No	25.82	Yes	No	No	0	30	No	Male	80 or older	White	Yes	Yes	Fair	8	No	No	No	
No	25.75	No	No	No	0	0	No	Female	80 or older	White	No	Yes	Very good	6	No	No	Yes	
No	29.18	Yes	No	No	30	30	Yes	Female	60-64	White	No	No	Poor	6	Yes	No	No	
No	34.34	Yes	No	No	21	8	Yes	Female	65-69	White	No	Yes	Fair	9	No	No	No	
No	31.66	Yes	No	No	5	0	No	Male	60-64	White	No	Yes	Very good	5	No	No	No	
No	24.89	No	No	No	1	0	No	Female	55-59	White	No	Yes	Very good	7	No	No	No	
No	36.58	No	No	No	0	0	No	Female	60-64	White	Yes	No	Good	5	No	No	Yes	
No	25.84	Yes	No	No	5	0	No	Male	70-74	Black	No	Yes	Good	8	No	No	No	
No	30.67	No	No	No	4	4	Yes	Female	80 or older	White	No	Yes	Fair	8	Yes	No	No	
No	45.25	No	No	No	30	0	Yes	Male	70-74	White	Yes	No	Good	8	No	No	No	

Filters used:

Aggregation functions: `COUNT([HeartDisease])` was primarily used for total case mappings, and `AVG([BMI])` was applied to monitor ongoing metabolic baseline shifts. Worksheets utilize coordinated dynamic action filters (e.g., clicking on a specific Age Bracket automatically updates the surrounding Stroke and Diabetic charts) to allow synchronized dashboard cross-filtering.

Logic / Syntax: Mapped via custom discrete aggregate formulas linked with demographic parameters (e.g., calculating percentages for Race-wise Heart Disease distributions).

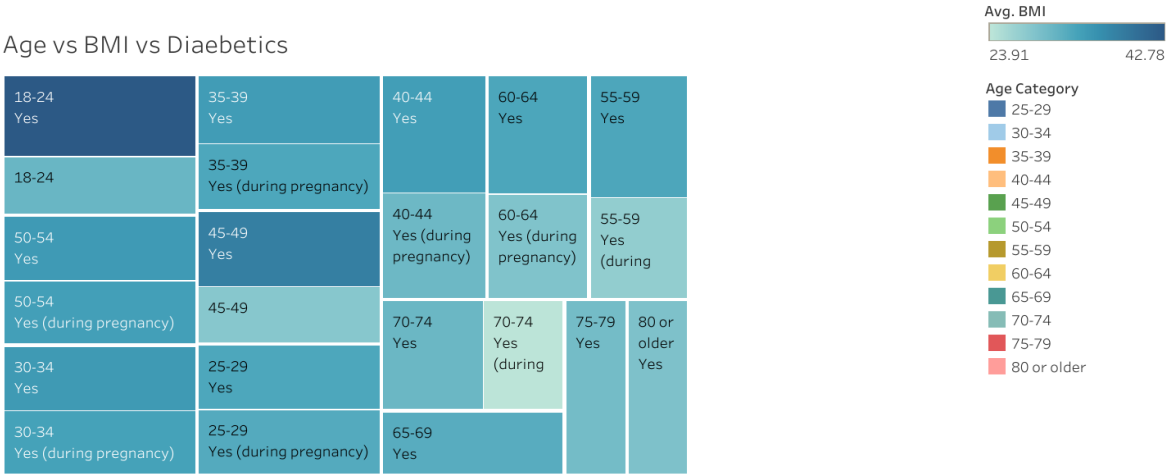
4. **Calculation Fields Used**

Purpose: Solves Tableau's order of operations conflict, allowing the dashboard to dynamically rank compound risk cohorts (e.g., patients with severe Diabetes AND a prior Stroke) within specific age-group cloud filters without breaking the background layout calculations.

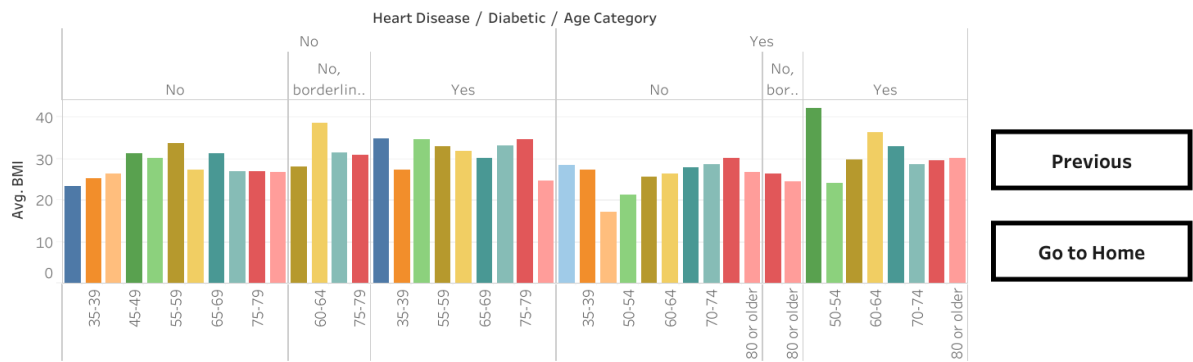
S.No.	Parameter	Screenshot / Values
		All visualizations are successfully embedded into a cohesive Dashboard Design template for clinical tracking.

Fig. -> Dashboard 1 (Demographics Matrix)

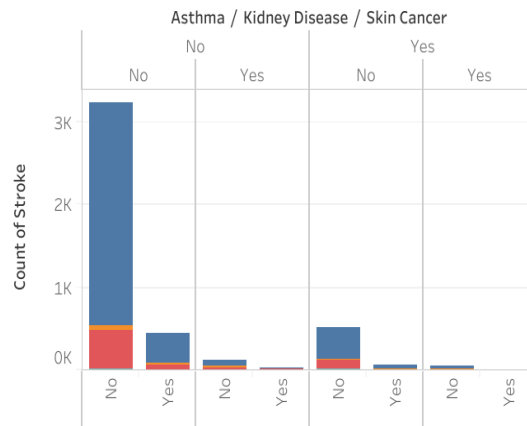
5.Dashboa
rd Design



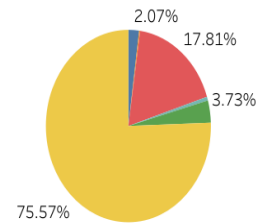
People Got Stroke Suffering from Diabetes and Heart Diseases



Other Health Diseases vs Stroke



Race Wise Heart Disease



Diabetic

- No
- No, borderline diabet..
- Yes
- Yes (during pregnanc..

Race

- American Indian/Alas..
- Asian
- Black
- Hispanic
- Other
- White

Count of Heart Disease

483

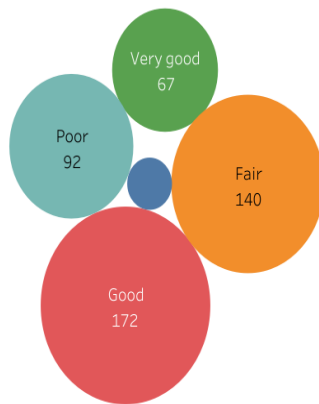
Gen Health

- Excellent
- Fair
- Good
- Poor
- Very good

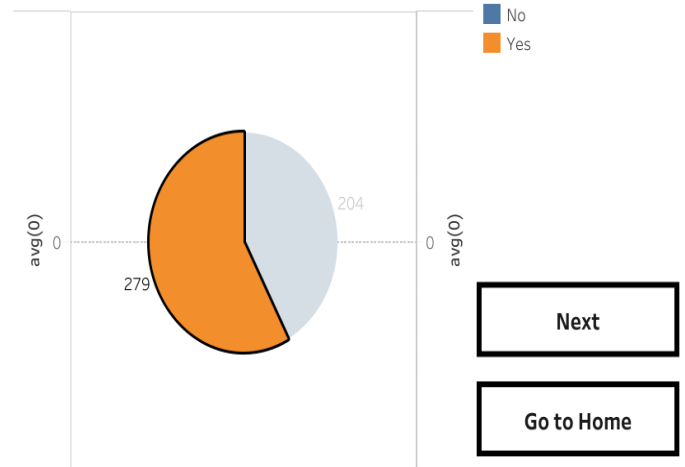
Physical Activity

- No
- Yes

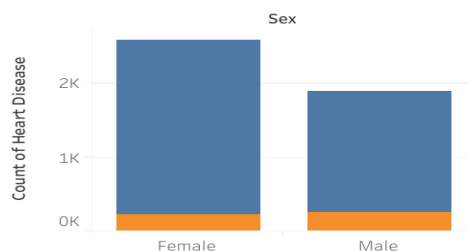
General Health vs Heart Disesease



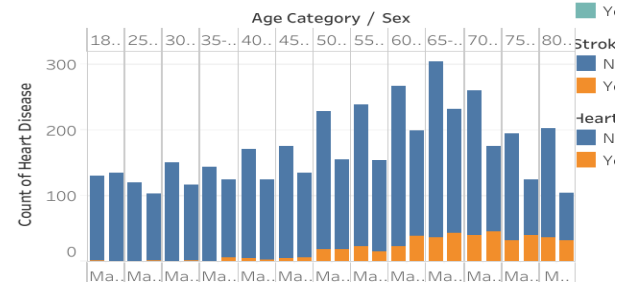
Physical Activity vs Heart Disesease



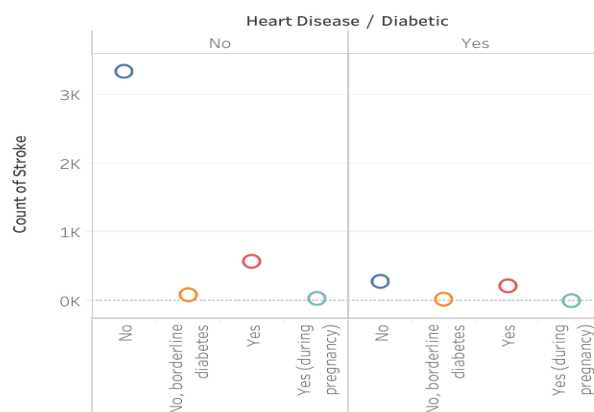
Gender vs Heart Diseases



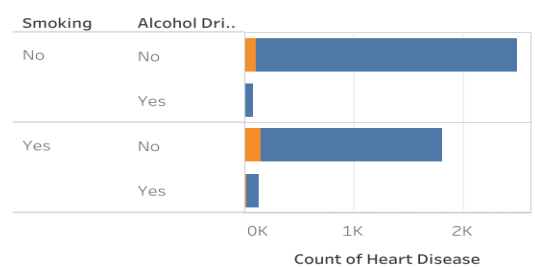
Age vs Heart Disease



Diabetic vs Stroke

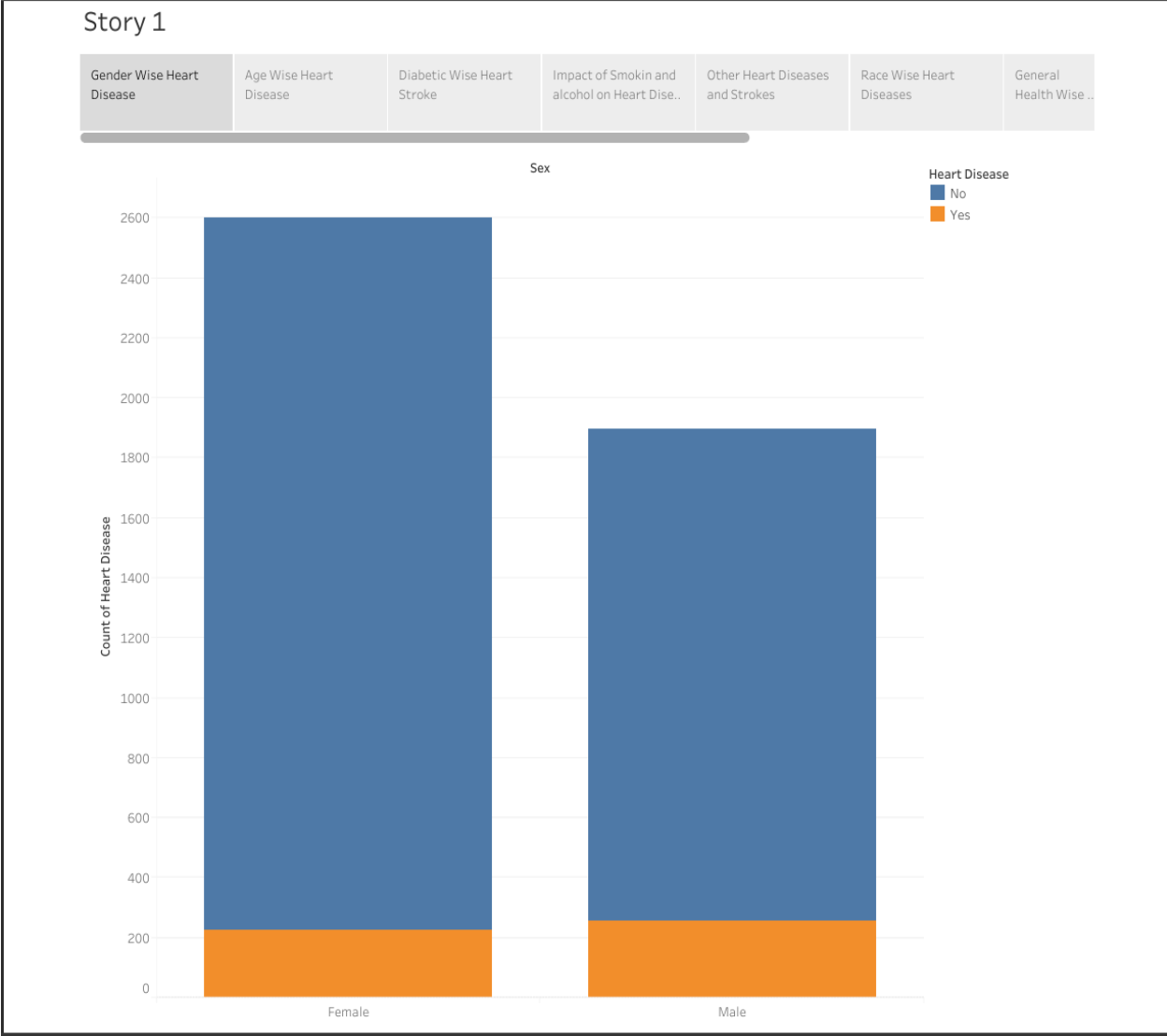


Impact of Smoking and Alcohol on Heart Disease



Heart Disease Analysis and Predictive Risk clustering utilizing Tableau Story. Multiple isolated clinical parameters and overlapping disease metrics are grouped into a sequential 7-slide Tableau storyboard.

Fig. -> Tableau Story Navigation Setup



6.Story
Design

Prepared as part of the Predictive Risk Clustering and Visual Informatics for Cardiovascular Disease capstone project.